

## SUMMARY TABLE: GENERAL SCIENCE GRADE 10

*Sub-Strand 1.6: Plant Growth and Development — Teacher Reference (Pre-filled)*

| SUMMARY TABLE: GENERAL SCIENCE GRADE 10 |                                                                                                                                                                                 |
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| <b>Sub-Strand</b>                       | Sub-Strand 1.6: Plant Growth and Development                                                                                                                                    |
| <b>Driving Question</b>                 | DRIVING QUESTION: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb? |

| INSTRUCTIONS                                                                                                                                                                                                                                                                            |
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| FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson. |

| Lesson # and Title                                                                            | What did I observe?                                                                                                                                                                                                                                                                                                                                                                      | What did I learn?                                                                                                                                                                                                                                                                                                                                        | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Lesson 1</b><br><b>Noticing and Wondering: Why Do Seeds Behave So Differently?</b>         | Students observed two transparent germination containers showing maize and bean seeds at day 3 to 5 of growth, noting that bean cotyledons arch above the soil while maize sends a single spike upward with the seed remaining underground. They also examined a photograph of a mango trunk cross-section showing growth rings and a photograph of a wilted two-season-old seed packet. | Students learned the distinction between growth — an increase in size or mass — and development — a series of organised changes in structure and function. They learned that scientists begin investigations by making careful observations and recording honest questions, as researchers at institutions like KALRO in Kenya do.                       | Students produced initial explanations for why maize and bean seeds emerge differently, drawing on their own prior knowledge and observations. These explanations are first drafts and are expected to be incomplete. Students also built Model Version 1 showing their current understanding of what is inside a seed and what it needs to grow, which will be revised across the sub-strand. |
| <b>Lesson 2</b><br><b>What Is Dormancy? Why Do Some Seeds Refuse to Wake Up?</b>              | We observed four seeds — fresh bean, soaked bean, new sorghum, and old stored sorghum — and noticed differences in seed coat texture, colour, and flexibility. We saw that soaked seeds felt softer and swollen compared to dry dormant seeds.                                                                                                                                           | We learned that seed dormancy has four causes: a hard impermeable seed coat, an immature embryo, chemical inhibitors such as abscisic acid, and unfavourable environmental conditions. We learned three methods of breaking dormancy: scarification, stratification, and soaking. We connected these to practices used by farmers in Turkana and Kisumu. | We can now explain part of the Kisumu farmer mystery: the stored seeds that did not germinate were in a state of dormancy — they were alive but their germination was blocked, possibly by a hardened seed coat or chemical inhibitors that built up during long dry storage. Dormancy is not death — it is a survival strategy that kept those seeds alive across two seasons.                |
| <b>Lesson 3</b><br><b>What Does a Seed Need? Investigating the Conditions for Germination</b> | We observed that maize and bean seeds placed on moist cotton wool in transparent cups showed no visible change on Day 1. We set up four experimental groups: one with full conditions, one without water, one                                                                                                                                                                            | We learned that three conditions are essential for germination: water (which triggers imbibition and enzyme activation), oxygen (which powers aerobic respiration to release energy for cell growth), and                                                                                                                                                | We explained that the Kisumu farmer is stored seeds that failed to germinate were likely in conditions similar to our Cup B — no water available. We also explained that a seed does not just need to have                                                                                                                                                                                     |

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|                                                                                                                | without air, and one in a cold location. We recorded our baseline observations and noted that on Day 1 all seeds looked similar regardless of the condition they were placed in.                                                                                                                                                                                                                    | optimum temperature (which ensures enzyme activity is efficient). We connected these conditions to the Kisumu farmer is garden: her seeds had water, air, and warmth, which is why most germinated. We also connected the No Water cup to the dry gourd storage from Lesson 2 — seeds in dry conditions cannot begin germination regardless of other factors.                                                                                                                                                                                                                                                                                                                                       | dormancy broken (Lesson 2) — it then actively needs three external conditions to be met simultaneously before germination can begin. Our experiment models this: removing even one condition is enough to prevent or delay germination, which is why the farmer is seeds respond so differently even when planted in the same garden on the same day.                                                                                                                                                                                                                                                                                                              |
| <b>Lesson 4</b><br><b>Two Ways to Emerge:</b><br><b>Epigeal and Hypogeal</b><br><b>Germination</b>             | Students observed germinating bean seeds with curved arch-shaped seedlings pushing yellow-green cotyledons above the soil surface, and germinating maize seeds with a straight green spike emerging above the soil while the grain remained buried underground. They noted that the bean seedling forms a distinct hook or arch shape while the maize seedling emerges as a straight upright point. | Students learned that the two types of seed germination are called epigeal (cotyledons emerge above the soil, driven by hypocotyl elongation — as in beans and groundnuts) and hypogeal (cotyledons remain below the soil while the epicotyl pushes the plumule upward — as in maize and wheat). They labelled diagrams with radicle, hypocotyl, epicotyl, cotyledon, and plumule, and completed a Venn diagram comparing both types.                                                                                                                                                                                                                                                               | Students explained the Kisumu farmer puzzle: the bean seedling forms an arch because the hypocotyl elongates and pulls the cotyledons above the soil where they can begin photosynthesis, while the maize seedling sends only its plumule upward via the elongating epicotyl, keeping the energy-rich cotyledon safely underground as a food reserve. Both strategies successfully get the seedling established, but through different structural mechanisms determined by the genetics of each plant species.                                                                                                                                                     |
| <b>Lesson 5</b><br><b>Getting Taller vs. Getting</b><br><b>Thicker: Primary and</b><br><b>Secondary Growth</b> | We observed cross-sections of a sukuma wiki stem and a mango twig. The sukuma wiki stem was green, soft, and showed loosely arranged vascular bundles with no visible rings. The mango twig was brown and hard, showed a distinct ring between the bark and the inner wood, and had visible growth rings in the woody core.                                                                         | We learned that plants grow in two distinct ways: primary growth increases the length of roots and shoots through cell division in apical meristems at the tips of roots and shoots; secondary growth increases the girth (thickness) of stems and roots through cell division in the vascular cambium, a ring of lateral meristem tissue located between the xylem and phloem. Each growing season, the vascular cambium of a Kenyan mango or avocado tree produces a new layer of xylem inward and phloem outward, creating an annual ring. Herbaceous plants such as sukuma wiki, beans, and maize have little or no vascular cambium activity and therefore do not develop a thick woody trunk. | We can now explain why the mango and avocado trees next to the Kisumu farmer bean plants grow thicker trunks every year while the bean plants remain thin: the trees have an active vascular cambium that produces new wood (secondary xylem) each season, while the bean plants rely almost entirely on apical meristem activity for growth in length and have no significant secondary growth. This also explains why counting the annual rings in a mango trunk cross-section can reveal the age of the tree, and why timber from Kenyan trees is dense and strong — it is the accumulated product of many years of secondary growth from the vascular cambium. |
| <b>Lesson 6</b><br><b>Chemical Messengers in</b><br><b>Plants: Growth</b><br><b>Hormones</b>                   | A sukuma wiki stem bending toward a window; a potted seedling tilted toward a one-sided light source; mangoes and bananas ripening quickly when stored together in a Kisumu market; dormant seeds that refuse to germinate even when given water and warmth.                                                                                                                                        | Plants produce five major hormones: auxins control phototropic bending and apical dominance by causing unequal cell elongation; gibberellins promote stem elongation and help break seed dormancy; cytokinins stimulate cell division and lateral bud growth; abscisic acid maintains seed                                                                                                                                                                                                                                                                                                                                                                                                          | The Kisumu farmer saw the bean seedlings push upward and arch while maize shot straight — differences shaped by germination type and apical dominance controlled partly by auxin. The dormant seeds that refused to germinate despite water and warmth were held by abscisic                                                                                                                                                                                                                                                                                                                                                                                       |

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|                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                  | dormancy and closes stomata during drought; ethylene triggers fruit ripening and leaf drop. These chemical messengers work by signalling specific cells to grow, divide, or stop, without the plant having a nervous system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | acid acting as a chemical lock. The mangoes in the market ripened faster near bananas because ethylene released by one fruit triggered ripening in the other. The spindly sukuma wiki near a dim window elongated toward light because auxin migrated to the shaded side, causing those cells to grow longer and bend the stem toward the light source.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Lesson 7</b><br><b>Factors Favouring Plant Growth: Environmental Influences and Cumulative Model Building</b> | We observed how tea plants in cool Kericho, maize in the warm Rift Valley, and mangroves in saline Mombasa each grow under very different environmental conditions — and that water, temperature, light, and nutrients shape not just whether a plant grows, but how it grows, how fast, and in what form.                                                                                                       | We learned that environmental factors do not act independently — they interact with the biological systems inside the plant, including the conditions needed for germination, the type of growth (primary or secondary), and the action of hormones such as auxins, gibberellins, and abscisic acid, to produce the plant forms we see in different regions of Kenya.                                                                                                                                                                                                                                                                                                                                                                             | We can now explain more completely why the Kisumu farmer sees such different outcomes: the maize and bean seeds differ in germination type and hormone responses, the mango tree undergoes secondary growth driven by cambium activity, and all of these biological processes are shaped and modified by the water, temperature, light, and nutrients available in the garden environment. Plants are not growing in isolation — they are growing in conversation with their environment.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Lesson 8</b><br><b>Putting It All Together: Synthesis, Cumulative Model Revision, and Assessment</b>          | We observed the complete evidence gallery from the unit — maize and bean germination in the Kisumu garden, the mango trunk cross-section with growth rings, spindly sukuma wiki grown in low light versus healthy sukuma wiki in full sun, germination data tables, and auxin redistribution diagrams — and we revisited the Driving Question Board to see every question our class has answered since Lesson 1. | We learned that a seed journey from dormancy to a mature plant is controlled by interacting biological and environmental systems: dormancy is maintained by abscisic acid and broken by water, temperature, and scarification; germination type — epigeal in beans and hypogeal in maize — is determined by the position of cotyledons; primary growth at apical meristems increases length while secondary growth at the vascular cambium increases girth; auxins, gibberellins, cytokinins, abscisic acid, and ethylene each regulate specific stages; and environmental factors including water, temperature, light, and nutrients interact with these hormones to produce the plant forms seen across Kenyan regions from Kericho to Mombasa. | We explained the complete anchoring phenomenon: bean seeds in Kisumu undergo epigeal germination because the hypocotyl elongates and pushes cotyledons above soil, while maize undergoes hypogeal germination because the epicotyl elongates and the coleoptile emerges while cotyledons remain underground; stored seeds failed to germinate because dormancy was maintained by abscisic acid and unfavourable conditions; and the mango trunk widened over the years through secondary growth driven by the vascular cambium. We also explained the Nakuru farmer phenomenon — pale, spindly bean seedlings result from etiolation caused by auxin redistribution in low light, which promotes excessive primary growth in length while suppressing lateral branching through apical dominance, and the solution is to increase light exposure or remove the apical bud to break apical dominance. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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